

Remove the Target. Preserve the Others.

Research focus: selective object removal when target and protected content interact.

The practical problem

Remove the specified instance and its effects.

Preserve nearby faces, text, structure and other instances.

A plausible image can still be an incorrect edit.

Our current stage

Editing infrastructure is implemented.

Strong-baseline validation is in progress.

A new method contribution remains to be demonstrated.

What We Have Built

Implemented pipeline; active denoising intervention remains a proposal.



Contract: target, operation and preservation requirements.

Controls: source-only masks and protected-region subtraction.

Final compositing preserves pixels outside the mask; it does not prove better generation.

Progress and Evidence

Status snapshot from the September 15 progress session; live jobs must be checked separately.

Verified progress

30 unit tests passed across spatial masks, executor and contract schema.

Offline LoRA loading fixed.

15 pilot configurations passed input preflight.

Execution status

Job 1042473 submitted; last session check: pending.

ObjectClear: code and weights ready, pipeline import unresolved.

No new generated results were available at that check.

Where a Contribution Must Go Further

A contract, verifier or final composite alone is not the proposed contribution.

ObjectClear: target/effect localization and background fusion.

Consistency Critic: reference-guided consistency correction.

AdaEraser: adaptive attention suppression for removal.

Hypothesis: explicit instance ownership can reduce collateral damage when effects or boundaries overlap.

Sources: arxiv.org/abs/2505.22636v2 | arxiv.org/abs/2511.20614 | arxiv.org/abs/2605.15921

Candidate Mechanism: Test Before Building

Proposed diagnostic, not a validated method or physical causal model.

1. Identify target, protected instances and uncertain regions.
2. At the same denoising state, vary target and protected reference evidence separately.
3. Test whether prediction responses distinguish target persistence from protected-content damage.
4. Build a regional controller only if this signal beats simple spatial heuristics.

Compare equal information and compute budgets. Report raw outputs before compositing.

Next Experiments and Decision Criteria

Use existing cases first; no new benchmark is required for this diagnostic phase.

Experimental sequence

Run ICDIT: 5 conditions x 3 fixed seeds.

Validate ObjectClear, including fusion on/off.

Add strong removal comparisons and held-out cases.

Then test the smallest intervention module.

Continue only if...

The failure persists on strong baselines.

Removal improves at comparable protection damage, or vice versa.

Gains survive raw-output and held-out evaluation.

Simple masks and equal-cost resampling cannot explain the gain.